Parth Shah

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EDUCATION

UNIVERSITY OF CALIFORNIA, DAVIS
BS in Computer Science and Engineering
Class of '23 | Davis, CA
GPA: 3.7 / 4.0

SKILLS

Proficient with:
• Python • Java • C++ • TypeScript • LangGraph • Gemini & Ollama • MCP • FastAPI • ReactJs • NodeJs • Git & Docker • AWS • Bash • LaTeX • SQL

AWARDS

2026 2nd AgentX - AgentBeats
2024 Bias for Action - Amazon
2023 Best Use of GitHub - HackDavis
2022 Scholars Award - Chevron
2022 1st - HackDavis
2022 Best Local Hack - SacHacks
2021 2nd - HackDavis
2020 Top 30 - PennApps XXI
2020 Top 20 - Google Solution
2020 Best Social Good - HackDavis

EXPERIENCE

AMAZON | Software Development Engineer 2
April 2025 - Present | Sunnyvale, CA

- Leading a cross-org effort to automate ticket investigation with AI agents. Live for 3 teams (30+ engineers) with 90% accuracy on holdout tickets, cutting root-cause time from 30-60 min to ~10 min across 1,200+ tickets a year.
- Built the agents' context from three layers: operational memory (past tickets clustered into reusable skills), a semantic code graph, and a knowledge layer of wikis, code reviews and design docs.
- Rewrote the seller contribution pipeline end to end to support LLM-based validation and enrichment, cutting CI/CD build times by 80%+. One of the core projects behind World Catalog, which powers Shop Direct: 250M listings for products sold on other sites, shown in Amazon search and Alexa.
- Launched 3 microservices with a plugin-based rules engine running 200+ rules for data validation and image enrichment.

AMAZON | Software Development Engineer 1
July 2023 - April 2025 | Sunnyvale, CA

- Architected the Progressive Listing Experience, using LLMs to turn a seller's product URL into an Amazon listing.
- Built and maintained Amazon's catalog contribution ingestion pipeline.

AMAZON | SDE Intern, Item Pipeline
June 2022 - Sep 2022 | Sunnyvale, CA

- Built a Code Analyzer that catches bugs and enforces coding standards before deployment.
- Rolled it out to the CI/CD pipelines of 8 teams, used by 100+ engineers.
- Built serverless microservices for the Item Pipeline with Java, TypeScript, AWS Lambda and SNS.

OPAL, LLC. | Junior Developer
Sep 2021 - June 2022 | Remote

- Built an NFT marketplace with Flask, Django and React.
- Led the social media team, building a platform for users to share digital assets.

PROJECTS

KIWI | Paper
July 2026 - Oct 2026

- Built Kiwi's decentralized inference engine, which splits an LLM across a swarm of small machines and streams activations between them peer-to-peer over libp2p.

- Handles machines joining and dropping out with VRAM-aware layer assignment, heartbeat-based eviction, rerouting and paged KV cache management.
- Built decentralized LoRA fine-tuning on top of it as part of the team (43% faster on 7 workers, instant recovery when a worker drops). Published at the NeurIPS 2026 CODEC-FM workshop.

NAAMSE | Website | GitHub | Paper
Jan 2026 - Present

- Built the core of an evolutionary AI red-teaming framework that uses genetic algorithms to mutate prompt-injection attacks.
- 2nd place out of 1,500+ submissions at Berkeley RDI's AgentX competition. Later forked by Mozilla's 0din team.

HASHIRU | GitHub | Paper
Jan 2026 - March 2026

- Built a budget-aware multi-agent framework where a 'CEO' agent manages specialized 'employee' agents.
- Keeps compute, inference cost and token usage within budget across parallel agents.

RUBATO | Live | GitHub
Nov 2026 - Present

- Built a real-time chord recognizer that runs entirely in the browser: a streaming CNN + Transformer names 108 chords from live audio every 40 ms.
- Trained on Billboard and Bollywood songs, reaching 83.4% major/minor accuracy on generated chord sheets, with smoothing that adds no latency.

GEM5 VISION | Website | GitHub | Poster
Jan 2023 - June 2023

- Built search and categorization for the gem5 Resources site, making simulator resources easier to find.
- Added support for multiple database backends for storing and serving resources.

STADIA WIRELESS | GitHub
July 2022

- Made the Google Stadia controller work wirelessly on any device using WebSockets.
- 450+ GitHub stars, covered by The Verge and Ars Technica.